CASE STUDY: FULLSTACK PRODUCT ENGINEER

AI Interview Platform — Monozukuri Product & Architecture Revamp

Candidate:	Fullstack Product Engineer	**Repository:**	github.com/rakamindev/ai-interview-platform
Feature Branch:	revamp/fullstack-product-engineer	**Submission Date:**	October 18, 2026
Claimed Depth:	Balanced Fullstack Slice (Backend Data Integrity & Mutex + Frontend UI/UX & UU PDP Compliance)	**PR Strategy:**	Option A: Comprehensive Single Pull Request

1. Executive Summary & Monozukuri Core

Transforming an unstable proof-of-concept into an enterprise-grade hiring platform

This assessment delivers an end-to-end revamp across both the Rails 7 backend and the React 18 / Vite frontend. Monozukuri represents our founding benchmark: software crafted with deep empathy for candidates and assessors, uncompromised data integrity across failure states, and pixel-level UI Polish.

#	Area	Impact & Shipped Solution
1	P0 Contract Fix	Resolved expected_level vs required_level disparity, ensuring the 'Required' column is accurately displayed across all Fit/Gap reports.
2	P0 Override Preservation	Re-engineered Portfolios::Generator to reconcile skills in-place during regeneration, eliminating destructive cascade deletion of human assessor overrides.
3	P0 Stampede Elimination	Replaced 404 polling storms with Redis in-progress locking (FitGapGeneratorWorker) and HTTP 202 Accepted status responses.
4	P1 Session Trap Fix	Enhanced Coverage::MapInjector#all_covered? so unprobed discovered skills in initiated state never trap candidates in indefinite interviews.
5	P2 Indonesian UU PDP	Built CandidatePDPNotice gating microphone access behind statutory consent for voice biometrics, AI evaluation, and human-in-the-loop oversight.
6	Testing Harness	Constructed comprehensive test suites (25 RSpec, 9 Vitest) and verified proven correctness via a Seeded Fault Test.

2. Deep Context: The Indonesian Talent Market & UU PDP

Regulatory compliance and human-centric design in talent evaluation

Hiring in Indonesia operates under distinct logistical realities: high applicant volume, geographic distribution across 17,000 islands, and severe recruiter cognitive overload. While asynchronous AI interviews present massive scalability leverage, they introduce critical vulnerabilities if built casually:

• **The Affected Person (Candidates):** Candidates do not choose an AI evaluation. An unpredictable session cutoff or a permanent state trap harms real lives. Software must provide absolute clarity, predictable timers, and transparent consent.

• **Indonesian Personal Data Protection Law (UU PDP No. 27/2022):** Enacted in late 2022, UU PDP classifies biometric voice recordings under Article 4(2) as *Data Pribadi Spesifik* (Specific Personal Data). Article 20 mandates explicit, voluntary, and informed consent prior to collection. Our statutory component (CandidatePDPNotice) fulfills this legal requirement before microphone activation.

• **Human-in-the-Loop Agency:** Under UU PDP Article 16, automated decision-making must allow human intervention. In Rakamin, human assessors can override any AI level with qualitative justification notes. Wiping those notes on portfolio regeneration violated both product trust and statutory principles.

3. Severity-Ranked Problem & Gap Analysis

Separating defective implementation from missing specification

Sev	Classification	Type	Real User Workflow Harm
P0	Fit/Gap Contract Mismatch	Defective	Backend returned expected_level while web expected required_level. The Required column was completely blank on the Fit/Gap report screen, hiding role baselines.
P0	Assessor Overrides Destroyed	Defective	Portfolio regeneration invoked destroy_all on portfolio_skills, triggering dependent: :destroy on assessor_overrides and wiping all human notes and rating corrections.

Sev	Classification	Type	Real User Workflow Harm
P0	FitGap 404 Polling Storm	Defective	GET /fitgap returned 404 while generating. Web caught 404 every 5s and queued duplicate Sidekiq workers, causing database connection pool exhaustion and duplicate calls.
P1	Discovered Skill Trap	Defective	MapInjector#all_covered? returned false if any discovered skill was initiated. Candidates were trapped in calls until hard time limit cutoff.
P1	Broken Skill Deletion in Forms	Defective	Removing a skill in Assessment or Vacancy edit omitted the item without sending _destroy: true. Rails ignored omissions, making skill deletion impossible.
P1	Overrides Omitted in Fit/Gap	Missing Spec	FitGap::Engine omitted is_override flags and notes in comparison payloads, meaning assessors could never see human override badges or rationale.
P2	Candidate Invite Port Mismatch	Defective	Session#invite_url pointed to localhost:3001 (API) instead of localhost:5173 (Vite web), resulting in 404 Routing Errors when candidates opened links.
P2	UU PDP Regulatory Non-Compliance	Missing Spec	No statutory disclosure or explicit consent gate for audio biometrics and AI scoring prior to microphone streaming.

TECHNICAL LEAD CONSTRAINT SIGNAL (ESCALATION):

In high-concurrency production deployments, free-form LLM JSON outputs risk schema drift. Furthermore, reliance on local Redis keys for worker locking should be standardized into a distributed Redis mutex (e.g. redlock-rb) with strict TTL fallbacks across all worker pools.

4. Solution Options & Architectural Trade-offs

Evaluating Option A (In-Place Reconciliation) vs Option B (Event-Sourced Ledger)

Evaluation Dimension	Option A: In-Place Reconciliation (CHOSEN)	Option B: Event-Sourced Ledger (REJECTED)
Product Impact	Immediate restoration of data integrity. Overrides remain intact across refreshes with zero schema breakage.	Full temporal audit trail of all score revisions, but introduces high UI complexity to display revision history.
Engineering Cost	Low to Moderate: Contained within Portfolios::Generator service with clean atomic transactions.	Very High: Requires event store tables, read projection workers, and CQRS synchronization.
Long-term Maintenance	High: Standard Rails ActiveRecord pattern; easily maintained by any junior/mid fullstack engineer.	Low: High cognitive burden; schema evolution across historic JSON event payloads requires complex upcasters.
Failure Modes	Gemini Pro timeouts handled gracefully by Sidekiq retries; in-place transaction rolls back cleanly.	Event projection lag; desynchronization between write events and read projections under load.
Contextual Codebase Fit	Optimal: Matches existing schema architecture without requiring risky data migrations.	Overkill: Disproportionate complexity for current stage of the platform.

Self-Derived Acceptance Criteria & Edge Cases Handled:
1. **Assessor Override Integrity:** If an assessor assigns L4 to a skill rated L2 by AI, regenerating the portfolio must preserve L4, keep notes intact, and update the underlying AI level to the new baseline.
2. **Candidate Trap Protection:** An initiated discovered skill must NEVER block interview completion if time remaining = 3 minutes or if no probe occurred within the last 2 minutes.
3. **Fit/Gap Contract Symmetry:** API returns both required_level and expected_level; frontend resolves c.required_level ?? c.expected_level safely.
4. **Nested Attribute Deletion:** Removing skills in edit views tracks deleted IDs and passes _destroy: true so records are permanently removed from Postgres.
5. **Idempotent Generation:** Duplicate POST /fitgap calls return HTTP 202 Accepted without re-queueing background workers.

5. Monozukuri Implementation & Testing Harness

Verified correctness across API, database, and client interface

We built complete automated test suites across both repositories to replace the previously empty test runners:

Service	Framework	Test Coverage Summary	Result
api/ (Backend)	RSpec 3.13 / Rails	FitGap::Engine comparisons, Portfolios::Generator override preservation, MapInjector trap guards, StateEngine rules, Session invite routing, Portfolios requests (200, 202, 404, deduplication), Assessment nested deletions.	25 Passed / 0 Failed (0.46s)
web/ (Frontend)	Vitest 5.0 / RTL	ComparisonTable contract fallback & override badges, CandidatePDPNotice statutory disclosures & consent toggle, SkillPortfolioCard effective levels & discovered tags.	9 Passed / 0 Failed (1.27s)

6. Seeded Fault Test Proof (Deliberate Mutation)

Proving tests catch real regressions rather than performing shallow green-checks

On a scratch branch (seeded-fault-test), we deliberately injected a contract fault in **FitGap::Engine** by corrupting required_level: nil and delta: 999. Running the RSpec test suite immediately caught the regression:

```
1) FitGap::Engine#call computes correct comparisons and satisfies frontend contract
Failure/Error: expect(react_comp['required_level']).to eq(3)
expected: 3
got: nil (compared using ==)
# ./spec/services/fit_gap/engine_spec.rb:105:in block (3 levels)
Finished in 0.20466s - 3 examples, 1 failure
```

The fault was reverted cleanly, returning the test suite to **25 passed, 0 failures**. This proves our test assertions actively protect API contracts against regression.

7. AI Verification Moment

Where AI code generation proposed unsafe patterns and human engineering intervened

THE RISKY AI SUGGESTION vs HUMAN MONOZUKURI CORRECTION:

When asked to ensure portfolio regeneration was idempotent, an AI code assistant proposed calling 'portfolio.portfolio_skills.destroy_all' prior to saving newly generated skills. While superficially idempotent, this ignored the

database relation 'has_one :assessor_override, dependent: :destroy'. Blithely adopting this would have caused catastrophic data loss in production. Human engineering intervened to design an In-Place Reconciliation algorithm that indexes existing overrides

in memory, updates existing skills in-place, and re-links overrides with zero data loss.

8. Video Demonstration & Defense Readiness

Demonstration link and preparation for CTO / Tech Lead live technical defense

• **Video Walkthrough Script:** A complete, minute-by-minute voiceover script and storyboard has been prepared in code>assessment/VIDEO_DEMO_SCRIPT.md/code>.
• **Walkthrough Content:** Problem discovery, Candidate UU PDP statutory consent flow, Assessor override workflows with persistence proof, Background job deduplication, and Test suite execution.
• **Live Technical Defense Readiness:** Ready to defend trade-offs, rejection of Event Sourcing (Option B), in-place transactional reconciliation, and distributed concurrency controls with the CTO and Technical Lead.